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TenderReady AI

TenderReady AI helps businesses understand complex tender and RFQ documents, identify key requirements and prepare more confidently for submissions using AWS AI services.

1. Executive Summary

TenderReady AI is an AI-powered procurement assistant designed to help South African small and medium-sized enterprises (SMMEs), start-ups and first-time bidders understand tender and Request for Quotation (RFQ) documents. It uses AWS cloud and AI services to extract key information and present it through a readable summary, a requirements checklist and document-based question-and-answer support.

The team will aim to deliver a working responsive web prototype within 24 hours. The minimum viable product (MVP) will focus on PDF upload, document extraction, AI analysis, checklist generation and questions grounded in the uploaded tender. Company-document readiness checking is a stretch goal.

2. Problem Statement

Tender and RFQ documents can be lengthy and complex. A business owner may need to search many pages to find closing dates, compulsory briefing sessions, eligibility criteria, supporting documents, pricing schedules, evaluation criteria and conditions that could lead to disqualification. For smaller businesses and first-time bidders, overlooking one mandatory requirement may affect a submission.

TenderReady AI addresses this challenge by using AI to organise procurement information into an understandable summary, an actionable checklist and a question-and-answer experience. The tool supports preparation but does not make procurement or legal decisions.

3. Proposed Solution and MVP Scope

The web application will let a user upload a tender or RFQ PDF, extract its text and relevant structured content, identify important requirements, and review the results in a clear interface. Users can inspect key dates, briefing details, contract duration, eligibility requirements, supporting documents, evaluation criteria, pricing instructions and potential disqualification conditions. Users can also ask questions answered using the uploaded document.

Included in the 24-hour MVP	Excluded / future enhancements
PDF upload and secure document storage	Broad support for every scanned/image/document format
Text extraction using Amazon Textract	Advanced OCR correction workflows for many document types
Amazon Bedrock tender summary and key fields	Automatic tender matching against a business profile
AI-generated requirements checklist	Full application submission or procurement decisions
Document-grounded Ask TenderReady Q&A	Company-document readiness comparison unless core flow is complete

4. Frontend Design Specification

The frontend should be clear, responsive and usable on desktop and mobile-sized screens. The design concept uses navy and blue surfaces, white content panels and green status accents for a professional procurement-support experience. The interface should minimise jargon and keep the key actions visible. Every summary and checklist must be clearly labelled as AI-generated and must remind users to verify requirements against the original tender document.

4.1 Login / Sign-up. Provide a simple entry page with TenderReady AI branding and sign-in/sign-up actions. Amazon Cognito can provide authentication if included in the MVP.

4.2 Home / Tender Upload. Show a prominent PDF upload area, supported-format note, upload status and recent tender list. The upload workflow should clearly show the selected file.

4.3 Analysis in Progress. Show progress for upload, text extraction, AI analysis, and summary/checklist generation. Status indicators should reflect actual backend processing states.

4.4 Tender Summary. Display tender title and reference, closing date/time, compulsory briefing information, contract duration and other key fields extracted from the document.

4.5 Requirements Checklist. Present tender-specific submission requirements, such as company registration, tax documentation, B-BBEE documents, pricing schedules, references, signed declarations and tender forms where required.

4.6 Ask TenderReady. Provide a question-and-answer panel with suggested questions about briefing sessions, required documents, company experience, evaluation criteria, possible disqualification conditions and key dates.

4.7 Document Readiness — Stretch Goal. Compare detected supporting-document categories against tender requirements and identify possible gaps. A completeness percentage must not imply tender qualification or award probability.

4.8 Frontend Technologies

Technology	Purpose
React.js	Build reusable upload, analysis, summary, checklist and Q&A screens.
TypeScript	Improve code reliability and maintainability.
HTML5 and CSS / Tailwind CSS	Provide structure, responsive layouts and consistent styling.
AWS Amplify Hosting (optional)	Potential AWS-managed hosting route for the frontend.
Git and GitHub	Support source-code management and team collaboration.

5. AWS Architecture and Service Roles

The architecture uses managed and serverless AWS services to keep the 24-hour prototype achievable. The core workflow is: upload a PDF, extract its content, analyse the extracted text, store the results, and display a summary, checklist and Q&A interface. Optional features should only be added after the core workflow works.

AWS service	Role in the prototype	Notes / MVP use
Amazon S3	Tender document storage	Store uploaded PDF files in private storage with appropriate access controls.
Amazon Textract	Document extraction	Extract text, tables and forms from tender documents.

Amazon Bedrock	AI analysis and Q&A	Generate structured summaries, requirements and answers grounded in the tender content.
AWS Lambda	Backend processing	Validate requests and coordinate extraction, AI analysis and result preparation.
Amazon API Gateway	API layer	Expose endpoints for upload workflow, analysis retrieval and document questions.
Amazon DynamoDB	Metadata and analysis state	Store tender metadata, analysis results, checklist state and application records.
Amazon Cognito	Authentication	Provide sign-up/sign-in and associate a user's documents with their account.
AWS IAM	Access control	Apply least-privilege permissions between AWS services.
Amazon CloudWatch	Monitoring and logs	Capture logs and errors to support testing and troubleshooting.

5.1 End-to-End Workflow

The user signs in and uploads a tender or RFQ PDF through the web application.

The application stores the file in a private Amazon S3 bucket.

AWS Lambda coordinates the processing workflow and invokes Amazon Textract to extract text and structured content.

Amazon Bedrock analyses the extracted content and identifies key dates, mandatory requirements, supporting documents, evaluation criteria, pricing instructions and potential disqualification conditions.

The application stores analysis results and tender metadata in DynamoDB and displays the tender summary and AI-generated checklist.

The user asks questions; the application provides answers using the uploaded tender content as context.

The user verifies all information against the original tender before preparing a submission.

5.2 Privacy, Security and Responsible AI

Access controls. Use private S3 storage, appropriate encryption, least-privilege IAM permissions and account-level access checks.

Data privacy. Tender and company documents may contain sensitive business information. Avoid putting document contents in logs and configure suitable retention.

Evidence and transparency. Show source-page references where feasible and distinguish text extracted from the tender from AI-generated explanations.

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Responsible use. Treat summaries and checklists as decision support. Do not guarantee eligibility, legal compliance or tender success.

Future readiness. Before production, review privacy, security, access controls, retention, model behaviour and expected operating costs.

6. AWS Pricing Calculator Cost Estimate

The following is a preliminary planning budget for a low-usage development/demo environment. It is not an official AWS Pricing Calculator export. Assumptions: one small environment, a small group of test users, a limited number of tender PDFs per month, and a modest number of AI summaries and questions. Actual cost depends on AWS Region, document volume and page count, extraction features, selected Bedrock model, token volume, storage, API usage, logs and taxes.

Service	Main cost driver	Indicative monthly range (USD)
Amazon S3	Storage and requests	$1–$5
AWS Lambda	Invocations and execution duration	$0–$5
Amazon API Gateway	API requests	$1–$10
Amazon DynamoDB	Reads, writes and storage	$1–$5
Amazon Cognito	Monthly active users and selected features	$0–$5
Amazon Textract	Pages and processing features used	$5–$30
Amazon Bedrock	Model and input/output token volume	$5–$50
Amazon CloudWatch	Logs, retention and monitoring	$1–$10
AWS IAM	Access control	$0 additional service charge for this use

Indicative monthly planning range: approximately US$14–US$120. This range is indicative only and is not a confirmed AWS quote. Textract and Bedrock are expected to be the main variable cost items at prototype scale.

6.1 How to Generate the Required Official Export

Open the AWS Pricing Calculator at https://calculator.aws/ and select Create estimate.

Add Amazon S3, AWS Lambda, Amazon API Gateway, Amazon DynamoDB, Amazon Cognito, Amazon Textract, Amazon Bedrock and Amazon CloudWatch.

Select the intended AWS Region and confirm that the required Textract APIs and Bedrock model are available there.
Enter expected document volume and average page count, S3 storage and requests, API calls, Lambda runtime, DynamoDB usage, Cognito users, Bedrock model and estimated input/output tokens, and CloudWatch logs/retention.
Review monthly and 12-month totals and export the estimate as PDF or CSV.
Attach the AWS-generated file to the application and retain the estimate link for reference.
Official export instructions: https://docs.aws.amazon.com/pricing-calculator/latest/userguide/export-estimate.html

7. Development Timeline and Key Milestones

Phase	Key activities	Deliverable / milestone
Project setup	Confirm MVP scope, roles, sample tender, repository, AWS Region, permissions and model access.	Environment and development plan ready.
Frontend foundation	Build sign-in placeholder, upload page, navigation and responsive UI components.	User can navigate to the upload workflow.
Upload and storage	Connect frontend to API/backend and store PDFs in Amazon S3.	PDF uploads work and return a document reference.
Document extraction	Connect Lambda and Textract; handle extraction status and errors.	Text extracted from a sample tender.
AI analysis	Connect Bedrock and generate structured tender fields and a summary.	Analysis results appear in the interface.
Checklist and Q&A	Build checklist and Q&A screens; persist results in DynamoDB.	Core MVP user journey works end to end.
Testing and security	Test invalid PDFs, missing information, long documents, access boundaries and AI failures.	Critical defects addressed; limitations documented.

Demo and handover	Rehearse the demonstration, explain architecture, export cost estimate and prepare handover.	Working prototype and application material ready.

Timeline note: this proposed 24-hour plan assumes that the team has AWS access and a sample tender ready at the start. If integration issues arise, prioritise PDF upload, extraction, summary/checklist and document-grounded Q&A before the company-readiness stretch goal.

8. Target Market, Deployment Intent and Sales Plan (for PDM Discussion)

8.1 Target Market

SMMEs and start-ups. Businesses responding to public- or private-sector tenders and RFQs.
First-time bidders. Owner-managed businesses without dedicated procurement teams.
Procurement consultants. Professionals who assist multiple businesses with document review and submission preparation.
Business-support organisations. Incubators, enterprise/supplier-development programmes and SMME support networks.

8.2 Deployment Intent

Begin with a controlled web prototype using sample or authorised tender documents. Demonstrate the upload-to-analysis workflow and collect feedback from a small group of SMMEs or procurement-support partners. A pilot should follow only after reviewing extraction accuracy, privacy, security, AI behaviour, cost and service availability. Production deployment requires additional review of data handling, access controls, retention and operational support.

8.3 Sales and Go-to-Market Plan

Phase 1 User discovery. Interview SMMEs, first-time bidders and procurement consultants to validate the biggest document-review challenges.
Phase 2 Prototype demonstrations. Demonstrate the upload, summary, checklist and Q&A workflow and collect feedback on usefulness and clarity.
Phase 3 Pilot partnerships. Approach SMME incubators, business-support organisations, procurement consultants and enterprise/supplier-development programmes.
Potential revenue model. Test a free limited tier followed by pay-per-document or monthly subscription options. Confirm cost per document and willingness to pay before fixing prices.
Acquisition channels. Use demonstrations, workshops, partner referrals, SMME events and targeted digital outreach.

Success measures. Track sign-ups, activation, documents analysed, checklist usage, repeat use, customer feedback, cost per document and conversion to paid use.

9. Expected Impact and Evaluation

TenderReady AI aims to reduce the time businesses spend manually searching procurement documents and help users identify important requirements earlier. It is particularly relevant to smaller businesses and first-time bidders that may not have dedicated procurement teams.

- Whether key dates and mandatory requirements are extracted correctly from sample documents.
- Whether the generated summary and checklist are clear and useful to a test user.
- Whether answers are grounded in the uploaded tender and avoid unsupported claims.
- Whether a user can complete the upload, review and Q&A workflow.
- Processing time and estimated AWS cost per document at prototype scale.

10. Final Demonstration Script

Show a lengthy sample tender or RFQ document.
Upload the PDF to TenderReady AI and show the processing status.
Demonstrate extracted closing date, briefing details and other key information.
Show the AI-generated tender summary and requirements checklist.
Ask TenderReady a question such as: “What documents do I need to submit?”
Show that the answer is based on the uploaded tender content.
If the stretch goal is completed, compare detected supporting documents with the tender requirements.